

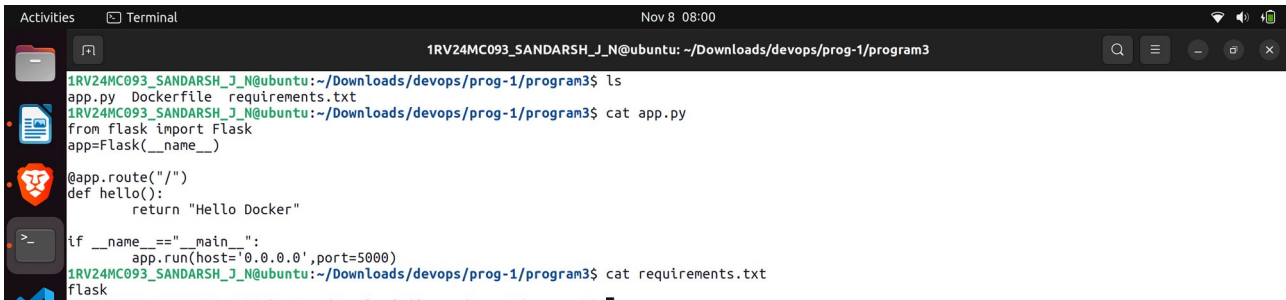
Program 3 : Code a dockerrized flask or node application

proj3

- app.py
- requirements.txt
- Dockerfile

Step 1:

app.py and requirements.txt

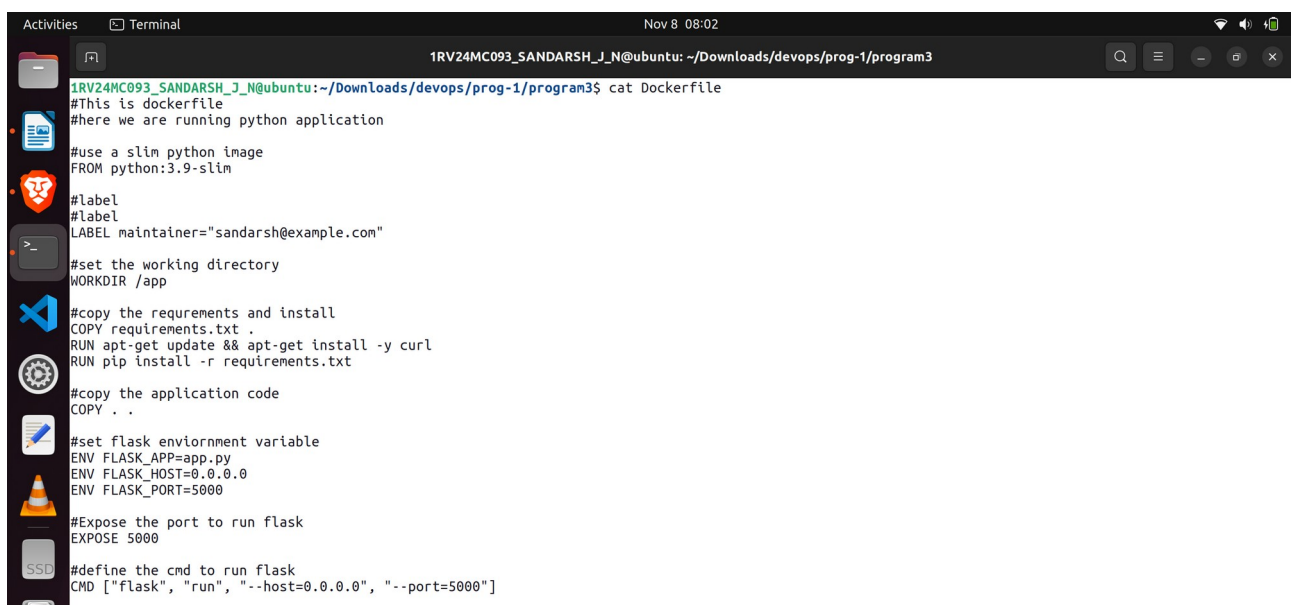


```
1RV24MC093_SANDARSH_J_N@ubuntu: ~/Downloads/devops/prog-1/program3
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prog-1/program3$ ls
app.py  Dockerfile  requirements.txt
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prog-1/program3$ cat app.py
from flask import Flask
app=Flask(__name__)

@app.route("/")
def hello():
    return "Hello Docker"

if __name__=="__main__":
    app.run(host='0.0.0.0',port=5000)
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prog-1/program3$ cat requirements.txt
flask
```

Step 2 : Dockerfile



```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prog-1/program3$ cat Dockerfile
#This is dockerfile
#here we are running python application

#use a slim python image
FROM python:3.9-slim

#label
#label
LABEL maintainer="sandarsh@example.com"

#set the working directory
WORKDIR /app

#copy the requirements and install
COPY requirements.txt .
RUN apt-get update && apt-get install -y curl
RUN pip install -r requirements.txt

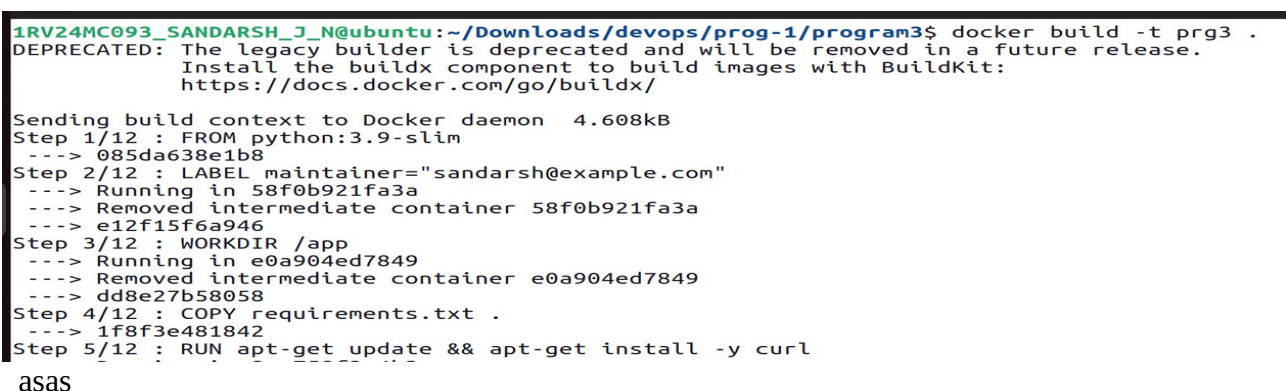
#copy the application code
COPY . .

#set flask enviornment variable
ENV FLASK_APP=app.py
ENV FLASK_HOST=0.0.0.0
ENV FLASK_PORT=5000

#Expose the port to run flask
EXPOSE 5000

#define the cmd to run flask
CMD ["flask", "run", "--host=0.0.0.0", "--port=5000"]
```

Step 3: build docker image



```
1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prog-1/program3$ docker build -t prg3 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.608kB
Step 1/12 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/12 : LABEL maintainer="sandarsh@example.com"
--> Running in 58f0b921fa3a
--> Removed intermediate container 58f0b921fa3a
--> e12f15f6a946
Step 3/12 : WORKDIR /app
--> Running in e0a904ed7849
--> Removed intermediate container e0a904ed7849
--> dd8e27b58058
Step 4/12 : COPY requirements.txt .
--> 1f8f3e481842
Step 5/12 : RUN apt-get update && apt-get install -y curl
asas
```

```

Step 7/12 : COPY . .
---> 476779dd266b
Step 8/12 : ENV FLASK_APP=app.py
---> Running in e2e7168b35b5
---> Removed intermediate container e2e7168b35b5
---> 5f1b54ccba3d
Step 9/12 : ENV FLASK_HOST=0.0.0.0
---> Running in 587fee508a34
---> Removed intermediate container 587fee508a34
---> 4064b49b77a6
Step 10/12 : ENV FLASK_PORT=5000
---> Running in 20748de9ec76
---> Removed intermediate container 20748de9ec76
---> 675620bd2f5e
Step 11/12 : EXPOSE 5000
---> Running in 530c3ae3c573
---> Removed intermediate container 530c3ae3c573
---> 2dd2c390c5d8
Step 12/12 : CMD ["flask", "run", "--host=0.0.0.0", "--port=5000"]
---> Running in 77cbd98b1819
---> Removed intermediate container 77cbd98b1819
---> 28d46fef23b6
Successfully built 28d46fef23b6
Successfully tagged prg3:latest

```

Step 4 : build a container

```

1RV24MC093_SANDARSH_J_N@ubuntu:~/Downloads/devops/prog-1/program3$ docker run -it --name prg--3 prg3
* Serving Flask app 'app.py'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [08/Nov/2025 02:39:12] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [08/Nov/2025 02:39:12] "GET /favicon.ico HTTP/1.1" 404 -

```

Step 5 : Output from browser

